WORKED SOLUTION

STEP I 2014, Question 1

Pure • Differences of Squares

> **STEP I 2014, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/14-s1-q1

The key idea

Everything rests on one factorisation: $a^2 - b^2 = (a-b)(a+b)$. Writing n as a difference of two squares is the same as writing $n = de$ with $d = a - b$ and $e = a + b$, and since d and e differ by the even number $2b$, they must have the same parity. Once you see that, every part becomes a question about factor pairs of the same parity.

Solution

Throughout, a and b are non-negative integers. If $n = a^2 - b^2$ with $n \geq 1$ then $a > b$, so we may always list a way as (a, b) with $a > b \geq 0$.

Part (i)

$$3 = 2^2 - 1^2, \quad 5 = 3^2 - 2^2, \quad 8 = 3^2 - 1^2, \quad 12 = 4^2 - 2^2, \quad 16 = 5^2 - 3^2.$$

(For 16 the obvious $4^2 - 0^2$ is not allowed, as the squares must be non-zero.)

Part (ii)

Consecutive squares differ by an odd number: $(m+1)^2 - m^2 = 2m + 1$. So any odd number $2m + 1$, with $m \geq 0$, is

$$2m + 1 = (m+1)^2 - m^2.$$

Part (iii)

Squares two apart differ by a multiple of 4: $(k+1)^2 - (k-1)^2 = 4k$. For $k \geq 1$ both $k + 1$ and $k - 1$ are non-negative, so

$$4k = (k+1)^2 - (k-1)^2.$$

For $k = 0$ we simply have $0 = 0^2 - 0^2$ (or $1^2 - 1^2$).

Part (iv)

Every square leaves remainder 0 or 1 on division by 4: $(2m)^2 = 4m^2$ and $(2m+1)^2 = 4(m^2+m)+1$. So $a^2 - b^2$ leaves remainder

$$0-0=0, \quad 1-1=0, \quad 1-0=1 \quad \text{or} \quad 0-1 \equiv 3 \pmod 4,$$

and never 2. Hence no number of the form $4k+2$ is a difference of two squares.

(Equivalently: if $(a-b)(a+b) = 4k+2$, then $a-b$ and $a+b$ have the same parity. Both odd gives an odd product; both even gives a product divisible by 4. Neither is $4k+2$.)

Part (v)

Suppose $pq = a^2 - b^2 = (a-b)(a+b)$ with $a > b \geq 0$. Put $d = a-b$ and $e = a+b$. Then d and e are positive integers with $d \leq e$ and $de = pq$. Conversely, any factorisation $pq = de$ with $d \leq e$ and d, e of the same parity gives exactly one way, namely

$$a = \frac{e+d}{2}, \qquad b = \frac{e-d}{2},$$

which are non-negative integers. So the ways of writing pq correspond one-to-one with such factor pairs (d, e).

Ask yourself Why do d and e always have the same parity here, and why are there only two pairs?

Since p and q are odd primes, pq is odd, so every divisor of pq is odd and the parity condition holds automatically. Take $p \leq q$. By unique factorisation, the only factor pairs with $d \leq e$ are

$$(d, e) = (1, pq) \qquad \text{and} \qquad (d, e) = (p, q).$$

(This is still true when $p = q$: the divisors of p^2 are 1, p and p^2.) These give

$$pq = \left(\frac{pq+1}{2}\right)^2 - \left(\frac{pq-1}{2}\right)^2 \qquad \text{and} \qquad pq = \left(\frac{p+q}{2}\right)^2 - \left(\frac{q-p}{2}\right)^2.$$

The two ways are distinct, because their values of $d = a-b$ are 1 and p, and $p \geq 3$. So pq is a difference of two squares in exactly two ways. For example, $15 = 8^2 - 7^2 = 4^2 - 1^2$. When $p = q$ the second way is $p^2 - 0^2$, which is allowed here, as only part (i) demands non-zero squares.

The result does *not* hold when $q = 2$. Then $pq = 2p$ with p odd, say $p = 2m+1$, so $2p = 4m+2$. By part (iv), $2p$ cannot be written as a difference of two squares in any way at all. (The factor pairs $(1, 2p)$ and $(2, p)$ each have one odd and one even member.)

Part (vi)

As in part (v), the ways of writing 675 correspond to factor pairs $675 = de$ with $d \leq e$ and d, e of the same parity. Since 675 is odd, all its divisors are odd, so every factor pair counts.

Now $675 = 27 \times 25 = 3^3 \times 5^2$, which has $(3+1)(2+1) = 12$ divisors. They pair off as $d \leftrightarrow 675/d$, and no divisor is paired with itself because 675 is not a perfect square ($26^2 = 676$). So there are $12/2 = 6$ pairs:

(d, e)	$(1, 675)$	$(3, 225)$	$(5, 135)$	$(9, 75)$	$(15, 45)$	$(25, 27)$
(a, b)	$(338, 337)$	$(114, 111)$	$(70, 65)$	$(42, 33)$	$(30, 15)$	$(26, 1)$

So 675 can be written as a difference of two squares in exactly **6** distinct ways.

Ask yourself Should (a, b) and $(-a, b)$, or $a^2 - b^2$ and $-(b^2 - a^2)$, count as different ways?

No. The question restricts everything to non-negative integers, and $a^2 - b^2 = 675 > 0$ forces $a > b$. So each way is a single pair (a, b) with $a > b \geq 0$, and these are what we have counted.

Checks

- Spot-check the table: $26^2 - 1^2 = 676 - 1 = 675$ and $30^2 - 15^2 = 900 - 225 = 675$.
- Parts (ii)–(iv) together say that $n \geq 1$ is a difference of two squares exactly when $n \not\equiv 2 \pmod 4$. A computer check of every n up to 2000 agrees, as does a check that pq has exactly two representations for all odd primes $p \leq q < 200$, and $2p$ has none.
- A search over all b finds exactly six representations of 675, the six in the table.

Common slips

- Forgetting the parity condition: counting a pair like $(2, p)$ for $2p$, which gives the non-integer $a = \frac{p+2}{2}$.
- Missing the case $p = q$ in part (v), or rejecting $p^2 - 0^2$ there. Zero is allowed outside part (i).
- Counting divisors (12) instead of factor pairs (6) in part (vi).

Beyond STEP

The pattern here, that the number of representations is controlled by the divisors of n, is typical of elementary number theory. A famous relative is Fermat's theorem on sums of two squares: an odd prime is $x^2 + y^2$ exactly when it is 1 mod 4, a statement proved neatly using the Gaussian integers $a + bi$. For more, read Davenport, *The Higher Arithmetic*.

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